Porter Cook

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Mechatronics engineer and Visiting Instructor of Practice at Utah Tech University with industry experience in automated test equipment, FANUC, Yaskawa, and Universal Robots programming, PLC-to-robot integration, and C#/SQL manufacturing software.

EDUCATION

B.S. Mechatronics Engineering – *Utah Tech University*, Saint George, UT **May '26**

Key Classes: Industrial Robotics, PLC Programming, Control Systems, Embedded Systems, Mechatronic System Design

SKILLS

Core: Industrial Robot Programming, PLC Programming, PLC-to-Robot Integration, Automated Test Equipment, Control Panel Design & Industrial Wiring, HMI Design, Troubleshooting & Commissioning, Technical Instruction, Lab Safety

PLC & Controls: Ladder Logic, Structured Text, Productivity Suite, Beckhoff TwinCAT 3, DirectLogic, SCADA, VFDs, Group I/O, Ethernet Communication

Robotics: FANUC (HandlingTool, RoboGuide), Yaskawa, Universal Robots, End-of-Arm Tooling

Software: C#, WPF, SQL, Python, C++, HTML

CAD / CAM: SolidWorks, Fusion 360, Onshape, AutoCAD Electrical, CNC Toolpaths & G-code

PROFESSIONAL EXPERIENCE

Visiting Instructor of Practice, Industrial Robotics **Apr '26 – Present**
Utah Tech University *Saint George, UT*

- Co-developed a new Industrial Robotics curriculum, adding robot motion principles (static and inertial forces), robot I/O, and Ethernet communication alongside robot programming and operation
- First instructor to bring FANUC RoboGuide into the course; students build, program, and debug virtual workcells before running physical robots, making hardware lab time more productive
- Co-designed a PLC-to-robot trainer pairing an AutomationDirect P1000 PLC with a FANUC LR Mate 200iD
  - Wired through CRMA58 terminal blocks for hands-on group I/O, digital logic, and Ethernet labs
- Established lab safety protocols for student operation of industrial robots; teach MTRN 2200 and MTRN 2205 covering safe industrial wiring and troubleshooting

Automation Technician **Mar '26 – Apr '26**
IME Automation *Hurricane, UT*

- Supported an automated system project from development through completion, handling controls integration and troubleshooting with end-of-arm tooling and robotic material handling

Automation Engineering Intern **Aug '24 – Sep '25**
Baxter Aerospace *Saint George, UT*

- Developed HVX, an automated wire-harness test system for Quality Assurance
  - Verifies continuity and performs high-voltage testing using PLC control and custom test sequences
- Built C# WPF applications backed by SQL databases to deliver MES work instructions to assembly lines
- Programmed Yaskawa, Universal Robots, and FANUC robots to automate manufacturing processes; integrated PLC control for sensor data collection and contributed to R&D on prototype automation projects

Makerspace Intern **Aug '22 – May '23**
Utah Tech University – Innovation Plaza *Saint George, UT*

- Lead operator for the Stinger CNC router (Fusion 360 toolpaths, G-code); installed E-stops and circuit breakers on shop machinery; trained users on laser engravers, CNC, and 3D printers

PROJECTS

Lovesac Material Dispensing System – *Senior Capstone, Industry Partner* *Utah Tech University*

- Designed and wired the main control enclosure (AC/24 VDC distribution, PLC I/O, dual VFDs with braking resistors) and wrote the ladder logic and HMI for the dispensing sequence
- Trained polynomial-regression and neural-network models (best RMSE 0.30 lb) to select motor settings
  - Optimized settings predicted within 0.05 lb of every 1–8 lb target weight

AR4 6-Axis Robotic Arm – *Systems Design*

- Designed and built a 6-axis arm on a Teensy 4.1 with a custom motor-control cabinet and PLC-triggered motion sequences

CERTIFICATIONS

FANUC Academy HandlingTool Operations & Programming (May '26) | Beckhoff TwinCAT 3 Getting Started (May '24)